<u>Week 13 – Capstone Progress Documentation</u>

Project ID: PW26_MG_02

Overview

Week 13 felt much like observing a trading system mature—from reacting to signals to truly understanding the market beneath them. What began as structured system design started evolving into something more intuitive, where each component no longer acted in isolation but responded as part of a larger, interconnected strategy. The transition toward a research-driven approach resembled shifting from basic indicators to deeper market insight—where patterns are not just detected, but interpreted with clarity.

The system gradually took the form of a layered trading architecture, where attention mechanisms behaved like adaptive traders focusing on critical signals, and spectral analysis worked like separating long-term trends from short-term noise. Optimization-based learning refined decisions much like risk-adjusted strategies in real markets. Every addition felt less like insertion and more like alignment—like tuning a system that was learning how to “read” the market.

Progress Update

This week’s developments mirrored the evolution of a disciplined trading strategy. Multi-head attention models began acting like multiple expert analysts, each capturing different dependencies within financial sequences. Orthogonal representations ensured that these “analysts” did not overlap in perspective, reducing redundancy and improving clarity—much like diversifying strategies to avoid correlated risk.

Spectral methods introduced a distinction between underlying trends and market noise, similar to separating signal from volatility in trading decisions. Temporal modelling, strengthened through gap-aware learning and smoothing, brought consistency across irregular data—like handling market gaps and discontinuities with stability. The inclusion of structured noise injection and uncertainty estimation reflected real-world trading conditions, where unpredictability is not avoided but accounted for. Gradually, the system transitioned from disconnected modules into a unified trading pipeline, where each component contributed to a coherent decision-making flow.

Upcoming Work

The next phase continues this journey toward a more refined trading intelligence. Multi-head attention will be extended to capture long-term dependencies, much like identifying macro trends beyond short-term fluctuations. Adaptive positional encoding will help the system better understand irregular market timing, reflecting real trading intervals. Sparse attention will introduce efficiency, ensuring the system scales like a well-optimized trading engine handling large volume of data. These components will be integrated into a unified pipeline and evaluated rigorously. At this stage, the process feels less like constructing a model and more like shaping a trading system that learns, adapts, and evolves—guided steadily toward clarity, precision, and deeper understanding.

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